Task 6: Sales Trend Analysis Using Aggregations

Objective

Analyze monthly revenue and order volume using SQL aggregations.

Dataset

The task brief specifies an `online_sales` table with:

• `order_date`

• `amount`

• `product_id`

• `order_id` (used for distinct order volume)

Tool

SQLite SQL is used for this submission. The same analysis can be adapted to PostgreSQL/MySQL by changing the date-extraction syntax.

Analysis Performed

1. Grouped sales by year and month.

2. Calculated monthly revenue using `SUM(amount)`.

3. Calculated order volume using `COUNT(DISTINCT order_id)`.

4. Sorted monthly results chronologically.

5. Added a query to identify the top 3 months by revenue.

6. Added an example query for filtering a specific time period.

Files

• `task6_sales_analysis.sql` — SQL queries for the analysis.

• `results_table.csv` — results table template/output when the dataset is available.

• `README.md` — project explanation.

Important

The task brief does not include the actual `online_sales` dataset rows, so numerical results cannot be calculated without inventing data. The SQL is complete and ready to run against the specified table.

Interview Questions — Short Answers

1. How do you group data by month and year?

Use date-extraction functions and `GROUP BY` year and month. In SQLite, `strftime('%Y', order_date)` and `strftime('%m', order_date)` can be used.

2. What's the difference between COUNT(*) and COUNT(DISTINCT col)?

`COUNT(*)` counts rows, while `COUNT(DISTINCT col)` counts unique non-NULL values in a column.

3. How do you calculate monthly revenue?

Use `SUM(amount)` and group the rows by year and month.

4. What are aggregate functions in SQL?

Aggregate functions calculate a value from multiple rows. Common examples are `SUM()`, `COUNT()`, `AVG()`, `MIN()`, and `MAX()`.

5. How to handle NULLs in aggregates?

Most aggregate functions ignore NULL values. `COUNT(*)` still counts rows, while `COUNT(column)` ignores NULLs. `COALESCE()` can be used when a NULL value needs to be replaced.

6. What's the role of ORDER BY and GROUP BY?

`GROUP BY` combines rows into groups for aggregation. `ORDER BY` sorts the final result.

7. How do you get the top 3 months by sales?

Group by year and month, calculate `SUM(amount)`, sort by revenue in descending order, and use `LIMIT 3`.